

Q4		
(a)	An oscillatory motion where the acceleration is directly proportional to displacement from equilibrium, and where acceleration is always opposite to displacement / acceleration is always directed toward equilibrium.	B1 B1
(b)	Since $a \propto -y$ By comparing with $a = -\omega^2 x$, $\omega^2 = \frac{\rho Ag}{M}$ $T = \frac{2\pi}{\omega}$ $= 2\pi \sqrt{\frac{M}{\rho Ag}}$ $= 2\pi \sqrt{\frac{0.012 + 0.025}{1000 \times 6.0 \times 10^{-4} \times 9.81}}$ $= 0.498 \text{ s}$ $= 0.50 \text{ s}$	B1 B1
(c)(i)	Energy of oscillation is the (maximum) kinetic energy the test-tube possesses, which decreases with time due to damping. Energy of oscillation = $\frac{1}{2}mv_0^2 = \frac{1}{2}m(\omega A)^2 = \frac{1}{2}m\omega^2 A^2$. A reduction of 75 % would mean that the energy of oscillation remaining is 25 % of its original. $\frac{E'}{E} = \frac{A'^2}{1.0^2}$ $\frac{1}{4} = \frac{A'^2}{1.0^2}$ $A' = 0.50 \text{ cm}$ From the graph, this happens at 1.0 s.	M1 A1
(c)(ii)	Natural frequency of the system is 2.0 Hz (since period is 0.5 s). However, the driving frequency is only 1.0 Hz. Energy transfer from the (external forcing agent) water waves to the test-tube is not optimal / does not result in resonance.	B1 B1

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